

17-Year Backtest of Straddles around S&P 500 Earnings Announcements

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Abstract

This paper investigates the efficacy of employing straddle derivatives strategies around S&P 500 earnings announcements to generate consistent alpha. Utilizing a comprehensive 17-year dataset, both long and short straddle positions are analyzed for their viability and profitability.

Long at-the-money (ATM) straddles initiated 30 trading days prior to earnings announcements displayed a decline in value leading up to one day before the event, primarily due to theta decay outweighing potential gains from rising implied volatility.

Conversely, a strategy involving buying a straddle one day before earnings and selling it one day after showed promising results, with a 108% compound annual growth rate (CAGR) and a Sharpe Ratio of 2.2 on 13,120 trades. However, risks such as single-week losses of up to 83.8% and unaccounted costs like commissions, slippage and taxes warrant caution and conservative trade-sizing.

As a counterparty position, short straddles near expiration significantly raised the risk of early assignment, rendering them unviable as a trading strategy. The paper advocates for a conservative approach to trading, emphasizing risk management and capital preservation by avoiding margin-enabled accounts to safeguard against potential losses.

Ultimately, the study advises against setting up long straddles 30 days prior to earnings or engaging in short straddles around earnings announcements as reliable means of generating alpha. Out of the money strangles and iron condors may be viable alternatives to short straddles but will require additional investigation.

Keywords: Straddle, Calls, Earnings Announcements, Equity Derivative Contracts, Equity Spreads, Extrinsic Value, Greeks, Implied Volatility, Intrinsic Value, Margin Call, Option Chains, Puts, Python, Quiet Period, S&P 500 Companies, SQLite

Introduction

Studies have shown that few traders can accurately time markets and consistently achieve positive expectancy with directional trades. In fact, Liu (2019) found that 89% of large-cap funds

underperformed the S&P 500 over a 10-year period up to 2019. In contrast, delta-neutral constructs have provided an alternative for traders seeking positive expectancy without calling direction. One such construct that has gained prominence is the straddle, sold short or bought long.

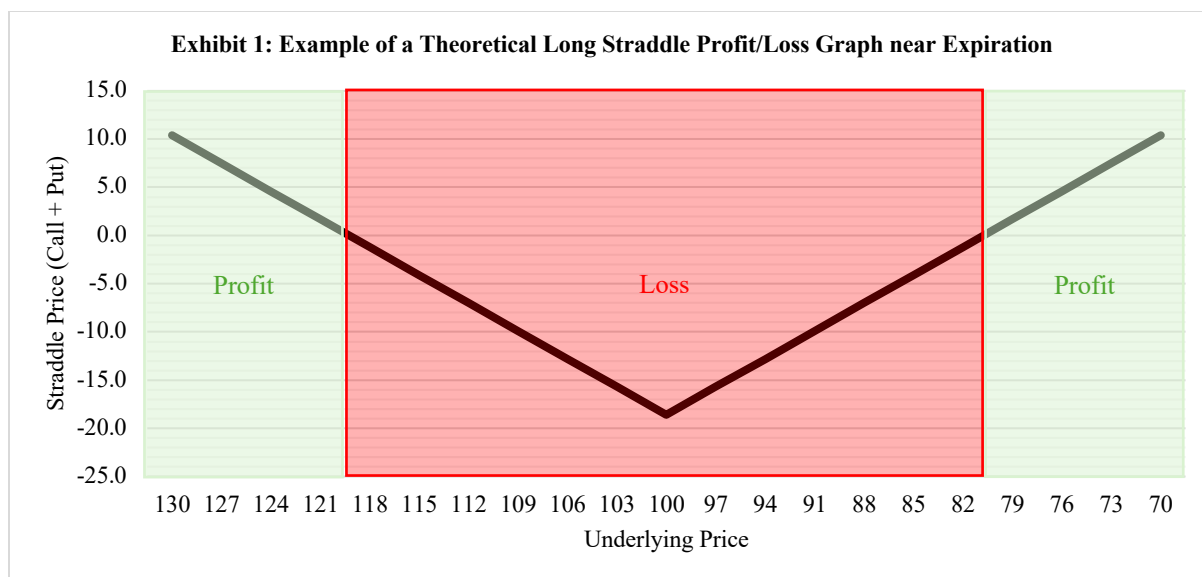
A long straddle is an undefined profit potential and defined risk construct (maximum loss equals total purchase price of each option, plus commission). It is initiated with the purchase of an ATM call and put option for the same underlying stock, strike price and expiration date. Irrespective of direction, this construct profits if there is significant price volatility in the underlying asset. If the underlying asset's price moves significantly in either direction, the value of one of the options increases, offsetting the loss on the other option and potentially resulting in an overall gain.

Long straddles are entered to speculate on near-term market-moving events such as earnings announcements, regulatory decisions, economic data releases or product launches, where uncertainty and volatility can spike. Long straddles are not profitable if they are range-bound where, at expiration, the sum value of the put and call does not exceed the initial purchase price. The key to managing risk is to calculate the breakeven point, where the combined cost of purchasing both options must be recouped to avoid losses.

Understanding straddles is best explained through an example (Exhibit 1). Assume that a trader buys a call and put option, costing \$9.30 each or a total of \$18.60 for ABC stock at \$100, before a significant event, followed by expiration (i.e., most of the straddle's value is intrinsic premium). Following the significant event, the investor profits if the stock price moves significantly above the breakeven points, which are calculated by adding the total premium from the strike price. The directional move—up or down—has to be large enough to overcome the summated premium of both options (i.e., \$18.60 + commission is the breakeven, not the individual option's cost of \$9.30). In this example, the underlying must move approximately +/- \$19 from the strike price by expiration to breakeven.

Because earnings announcements can trigger underlying asset volatility, pairing straddles constructs with earnings announcements has been suggested as a strategy with significant positive expectancy. In fact, Gao, Xing and Zhang (2018) calculated a 3.34% yield on ATM straddles entered 3 days before, leading up to the earnings announcement date.

This study will analyze both long and short straddle options when paired with earnings announcements. By gaining a deeper understanding of drivers of straddle performance, traders can determine if a straddle setup will be probabilistically profitable during earnings.



Methodology

Python was used to access S&P 500 company 10-Q earnings filings from the SEC via EDGAR's publicly available application programming interface (API) and subsequently mapped to option chains with the closest expiration date following each earnings announcement. 10-K filings were not used because the filing or report dates did not match actual company earnings announcement dates.

One period that the study delved into the behavior of straddles was during the "quiet period" mandated by the SEC for publicly traded companies. This period, typically lasting four weeks before the end of a fiscal quarter until the earnings report is released, prohibits the dissemination of insider information that could influence stock prices. The SEC aims to ensure fairness by preventing any advantage or disadvantage for investors. To comply, companies must follow specific rules for disclosing earnings reports, including issuing press releases, submitting forms to the SEC, and making information available on their website. During the quiet period, company executives typically refrain from giving interviews to financial journalists or analysts to avoid any perception of sharing insider information ("Quiet Period - Definition, Purpose, Reports").

To analyze this quiet period, the study used option chain slices of 30 trading days before earnings of each S&P 500 company, spanning 17 years of data from January 2007 through March 2024, acquired from Option Research & Technology Services (ORATS) as comma separated value files. Each daily file consisted of close prices that was ingested to construct a SQLite database table of approximately 1.7 billion rows.

Computation Assumptions

Daily option prices were computed as the mean of the bid and ask close prices. Since the goal of the study was to test the effect of earnings announcements on the straddle's price action, the straddle's close price one day prior to the day of the earnings announcement (earnings minus one day denoted as "E-1") was set as the reference denominator. Then all close prices before or after E-1 were divided by this reference denominator to generate a relative ratio that was then averaged across the cohort for each day as displayed in Exhibits 2-4. **Note:** the use of earnings day (E-0) as a baseline was challenging because pre/post market releases on earnings day varied across 17 years of data, distorting the signal-to-noise ratio.

Results

Our hypothesis posited that increased implied volatility (IV) leading up to earnings announcements would raise the prices of both the call and put options. Our results show that long ATM straddles on S&P 500 companies purchased 30 trading days prior to earnings experienced declining premium when held through one day before earnings announcements. The erosion of premium for ATM straddles due to theta decay was not offset by rising IV and because the underlying did not achieve a large enough pre-announcement move to drive either call or put option's intrinsic price higher.

Exhibit 2 plots straddle (call + put) prices relative to 1 day before earnings (E-1) and fits the data using a 2-degree, 3-term polynomial equation. Exhibits 3 and 4 break down Exhibit 2's price components into intrinsic and extrinsic values, respectively.

Within each Exhibit's polynomial equation, the coefficient of the second term reveals that this earnings-based ATM long straddle strategy cannot be profitable. The overall straddle shows a declining slope of -0.0154, because the intrinsic value's rising slope of 0.0111 cannot overcome the extrinsic value's rapidly declining slope of -0.0264. Furthermore, Exhibit 5 plots the rising mean IV, which is unable to contribute significant vega premium to the overall extrinsic value to offset rapid theta decay. Vega peaks at the money, requiring the underlying price to stay static for the 30-day holding period—a scenario that is refuted to the contrary by the rise in intrinsic value.

In contrast to pre-earnings, straddle prices stopped declining post-earnings through day 5 (Exhibit 2), despite a 8.2% decrease in IV from E-1 to E+5 (Exhibit 5). Intrinsic value constituted 27.0% of the straddle's value on E-30, rising to 88.4% on E+5 after earnings with the largest 2-day jump coming from one day before to after earnings: 72.4% to 86.6% of the straddle's price (Exhibit 3).

Exhibit 2: Mean Change in Straddle Call + Put Prices Relative to 1 Day Before Earnings

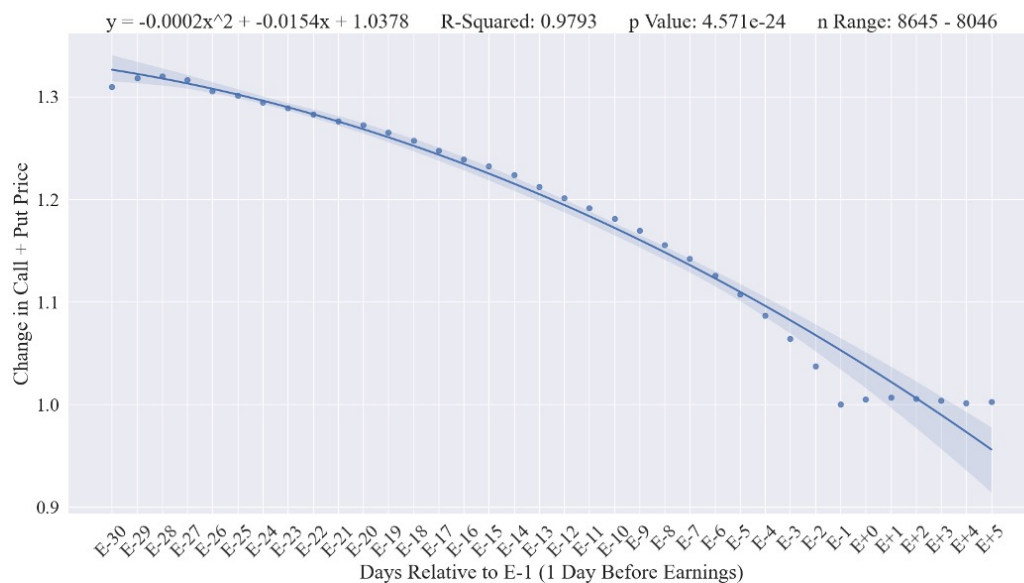


Exhibit 3: Mean Change in Straddle Intrinsic Value Relative to 1 Day Before Earnings



Exhibit 4: Mean Change in Straddle Extrinsic Value Relative to 1 Day Before Earnings

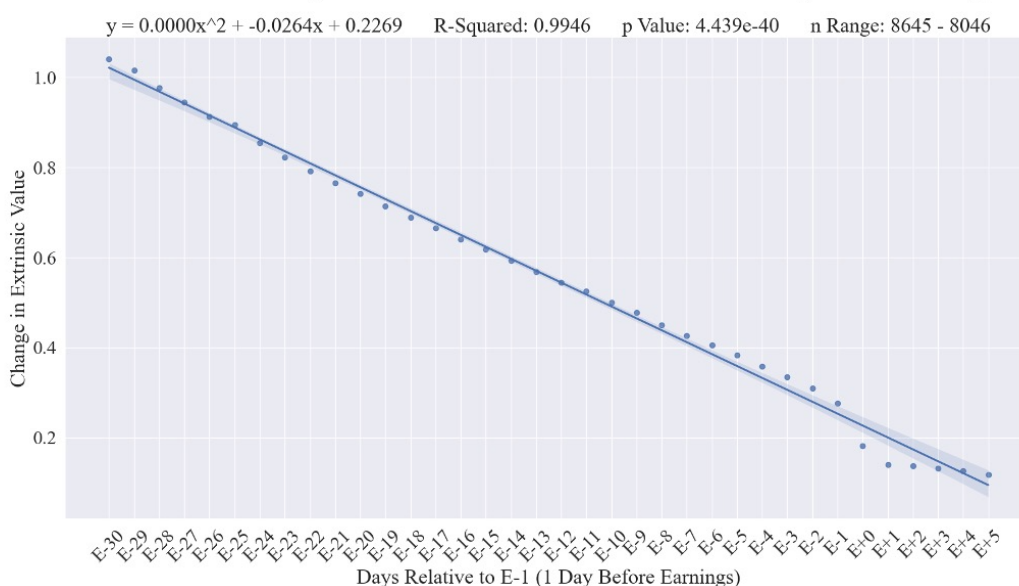
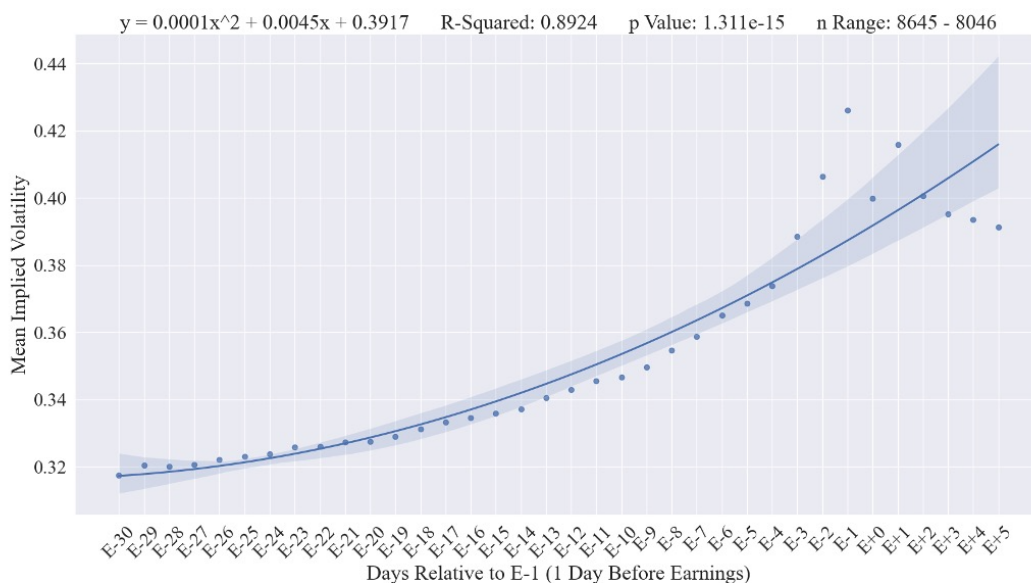


Exhibit 5: Rise in Implied Volatility through Earnings

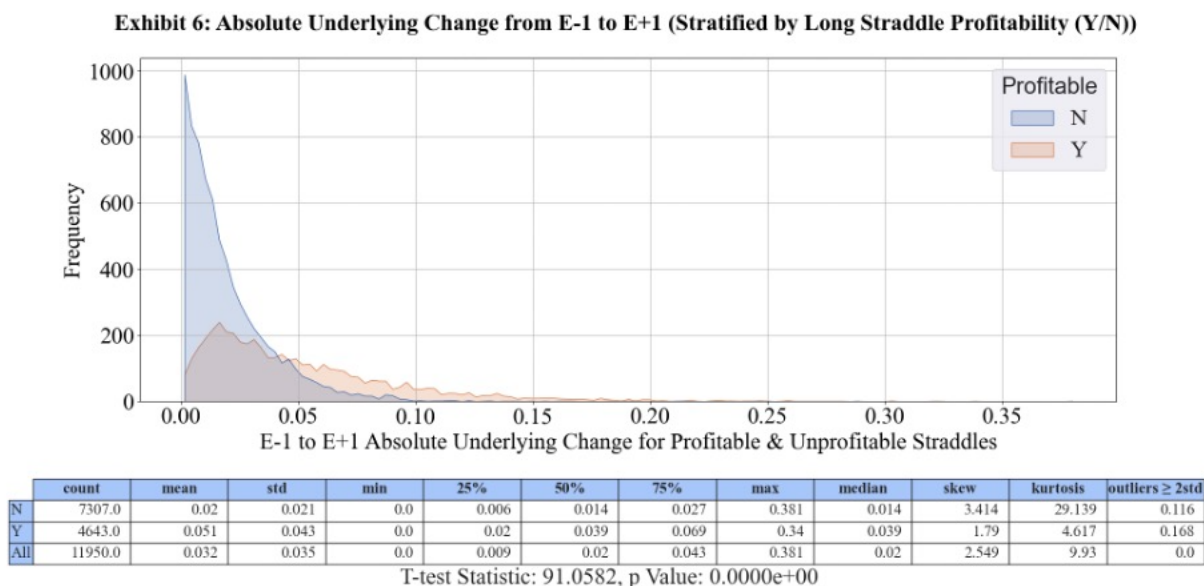


Next, we drilled down to study details of how straddle prices changed from one day before to one day after the release of earnings (i.e., traders would buy on E-1 and sell on E+1 all near market close), assessing whether the underlying price move could be large enough to drive positive expectancy. Beyond profiting from earnings volatility, this strategy also had an added advantage that capital would only be encumbered for 2 days instead of 30 days.

Long Straddles for E-1 to E+1 Durations

Exhibit 6 displays a positively skewed histogram plot of absolute underlying price changes and a table of descriptive statistics on profitable (Y), unprofitable (N) and combined data regarding the behavior of long straddles around earnings announcements. Long straddles displayed a profitable to unprofitable ratio of 0.39:0.61. However, profitable straddles demonstrated a significantly higher mean underlying price change of 0.051 ($\sigma=0.043$) from E-1 to E+1, while unprofitable ones showed a comparatively lower mean of 0.020 ($\sigma=0.021$). Median underlying price change, which was impacted less by outliers reinforced this finding with 0.039 vs. 0.014 for profitable and unprofitable straddles, respectively. This indicated that even across the middle range of observations, profitable straddles required significantly larger underlying moves relative to unprofitable straddles.

Significant kurtosis (i.e., “tailedness” and/or “peakedness”) of underlying price movement confirmed the marked presence of outliers in both profitable and unprofitable straddle distributions. For profitable straddles (Y), the kurtosis value of 4.617 signified a relatively high level of kurtosis, indicating heavy tails in the distribution. This suggested that there is a higher likelihood of extreme price moves—up or down—occurring during earnings announcements for profitable straddles. Similarly, the kurtosis value for all straddles (All) was 9.93, indicating a distribution with heavier tails and higher peaks (i.e., extreme underlying price moves were not uncommon). The highest kurtosis value (29.14) occurred for unprofitable straddles, because most of them made relatively small moves resulting in “peak” frequency closer to 0%.



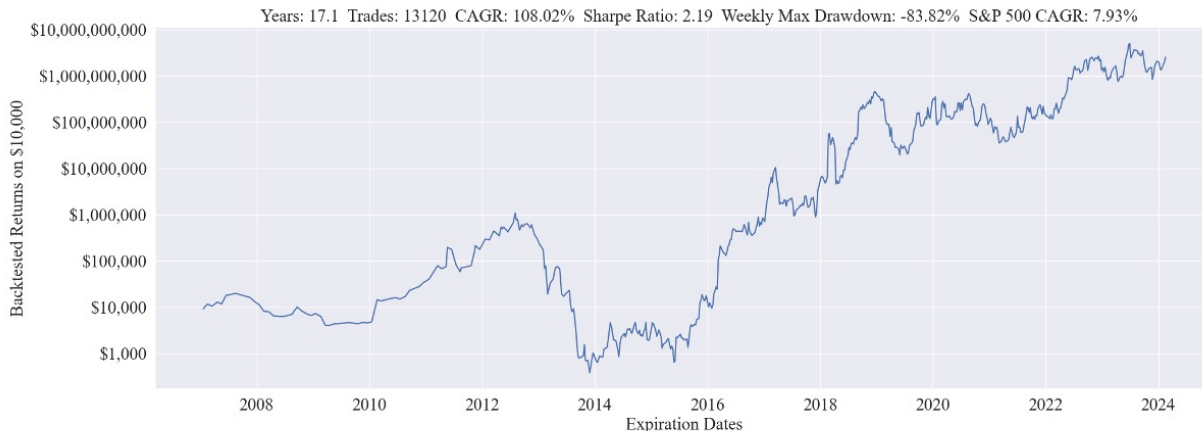
Based on these statistics, executing a long straddle strategy around earnings announcements may be more favorable if traders expect a significantly higher underlying price move than the mean (e.g., traders may consider a straddle if they expect a 1σ or greater move from the mean in either

direction). Of course, for this construct to be successful, traders must have accurate prognosticators to predict the magnitude of the underlying price's expected move.

To assess the positive expectancy “edge” without any prognosticators to predict profitable or unprofitable straddles, we performed a 17-year backtest on weekly expirations to impute the results of trading straddles on S&P 500 companies (Exhibit 7). Essentially, the idea was to be mechanical: can traders buy a straddle one day before earnings and sell it one day after earnings to generate positive expectancy?

Prior to the explanation, the authors advise caution on interpreting the theoretical results in Exhibit 7. First, this historical backtest excludes commission, slippage and taxes. The reader is advised to take these factors into consideration when interpreting the results.

Exhibit 7: Theoretical Returns of Long Straddles Held on S&P 500 Underlying from One Day Before to One Day After Earnings



	Trades	Probability	Mean Gain/Loss
Win	5408.0	0.412	0.629
Loss	7712.0	0.588	-0.269
All	13120.0	1.0	0.102

Second, the earlier years of the backtest had fewer companies with derivatives that expired in the week in which earnings were being reported. This simplified the mechanics of entering and exiting trades—an important consideration for retail traders. Over time, weekly expirations that included earnings releases became more frequent. In some cases, there were 237 earnings being released within the same week that their respective options were expiring. Executing hundreds of straddle trades per week while not challenging for institutional traders, may prove to be difficult for retail traders.

Third, the results do not address scalability considerations. While market makers are responsible for providing liquidity within the derivatives market, the analysis assumed that straddle trade tickets will be filled at the mean of the bid and ask price, irrespective of the number of contracts.

The purpose of the backtest was to assess the positive expectancy of long straddles. Over 17 years, the backtest included 13,120 trades that yielded a 108% CAGR with a Sharpe Ratio of 2.2. This is in comparison to the S&P 500 index return of 8% over the same period. The risk profile of this trade was extremely high because a single week alone incurred losses of 83.8%, so trading-sizing must be a critical consideration (e.g., 1% or less of the entire portfolio). Positive expectancy was observed with a win rate of 41% yielding 62.9% mean gain, while probability of losses was 59% with a mean loss of 26.9%. The resulting effect was a 10.2% gain on average per long straddle trade.

The E-1 to E+1 move had a 0.71% magnitude for straddles purchased on E-30, compared to 10.2% for those on E-1. This difference is due to the higher intrinsic value buildup for ATM straddles originated on E-30 by E-1, relative to the smaller buildup for those originated on E-1 (i.e., E-1 originations had smaller denominators). Of course, the larger loss in extrinsic value outpaced the intrinsic value buildup for E-30 originated straddles, which is why E-30 straddles could not yield positive expectancy.

Short Straddle

After analyzing long straddles, we switched focus towards the counterparty position of a short straddle for the E-30 to E+1 period. This altered the risk profile from defined to undefined risk (primarily due to the short call position), thus requiring a margin-enabled account.

The fundamental problem with this strategy near expiration is rooted in the risk of early assignment. At any given moment, either leg of the straddle will typically go in the money (unless the underlying does not move) and accumulate intrinsic value. Given the proximity to expiration, the counterparty is more likely to exercise the option, forcing the seller to cover the short leg (i.e., buy or sell the stock at the strike price).

Shorting straddles using inadequately capitalized margin-enabled accounts may trigger interest-based borrowing from the broker that can expose traders to additional costs and erode potential profits. Most importantly, margin-based borrowing can result in volatility-induced forced-liquidation that can trigger substantial losses (e.g., margin calls).

At expiration, several brokers will automatically assign shares for in-the-money options, so assignment of one leg of the straddle is imminent (“Understanding options assignment risk”). Thus, due to the substantial risk of assignment near expiration, short straddles were not deemed a viable strategy around earnings.

Early assignment risk can be mitigated by moving the straddle strikes significantly out of the money, effectively making the construct a strangle. As a result, several questions arise:

- Do strangles demonstrate significant positive expectancy around earnings?

- If positive expectancy is observed,
 - What is the optimal delta that maximizes CAGR and the Sharpe Ratio?
 - Because strangles are also an undefined risk construct, what does the outlier analysis reveal about worst case scenarios, specifically maximum drawdown situations?
 - To what extent can iron condor constructs that convert strangles to a defined risk profile mitigate maximum drawdown situations? What is the effect of converting strangles to iron condors on positive expectancy?
 - How long should strangles be held (i.e., up to one day before earnings, or through earnings when IV falls)?
 - Are there additional trade mechanics that can further optimize positive expectancy?

These questions, among others, are beyond the scope of this study, but can be examined as follow up research topics.

More broadly though, the authors advocate for a conservative approach to trading that not only avoids reliance on interest-based borrowing but eliminates it altogether. Because of the undefined risk profile of short straddles, even overcapitalized accounts may not fully safeguard against risks. That is, while the theoretical loss of a short put is limited to the underlying price atrophying to zero, there is no upper limit to the theoretical gain of an underlying stock, creating the potential for unlimited losses on short calls. To err on the side of caution, the authors' view is to disable margin-enabled accounts, thereby eliminating the risk of interest-based borrowing and forced liquidations.

Conclusion

In this study, we conducted a 17-year backtest of straddle options around S&P 500 earnings announcements, aiming to evaluate their performance as a consistent alpha-generating strategy. Our analysis looked at both long and short straddle positions, seeking to provide insights into their viability and profitability.

Our findings revealed that long ATM straddles initiated 30 trading days prior to an earnings announcement exhibited a decline in value when held until one day prior to the earnings event. Despite the anticipated increase in IV leading up to earnings, the erosion of premium due to theta decay outweighed any potential gains from rising IV, as the underlying stock failed to exhibit significant pre-announcement price movement. Consequently, the long straddle strategy did not yield positive expectancy for the E-30 to E-1 holding period.

Our analysis of buying straddles one day before earnings and selling it one day after showed promising results: 108% CAGR, 2.2 Sharpe Ratio over 13,120 trades, beating S&P 500 returns. However, risks were high, with single-week losses up to 83.8%. While the strategy's expectancy

was positive at 10.2% per trade, caution is advised due to unaccounted costs like commissions, taxes, slippage and scalability issues.

We saw early assignment near expiration as a significant risk factor when exploring the feasibility of short straddles as a counterparty position. The undefined risk profile, coupled with the potential for margin-induced forced liquidation, rendered short straddles around earnings unviable as a trading strategy. Examining strangles and iron condors where strikes are moved out of the money may be explored as a follow up research topic.

Our study underscored the importance of a conservative approach to trading, emphasizing risk management and capital preservation. We advocate for the avoidance of margin-enablement, as even overcapitalized accounts may not fully mitigate the inherent risks associated with short straddles. By prioritizing prudent trading practices and eliminating exposure to interest-based borrowing and forced sales, traders can safeguard their portfolios against potential losses and preserve capital. We do not recommend setting up E-30 to E-1 long straddles nor do we recommend short straddles around earnings announcements as a reliable means of generating alpha.

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